

Five Results and a Located Break

The Two Bills, the Forced Selector, the Novelty–Capacity Separation, a Negative Result on Order Dimension, and Why an Interface Costs Infinity

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Every number is quoted verbatim from live computation. Fifteen errors were made and corrected during this work; they are listed and classified in Section 8, because the dominant class turned out to be procedural rather than mathematical, and the procedural fix is the result with the longest reach. Classical results used as instruments are marked as such. Nothing here claims to derive an interface; Section 7 states precisely what was derived and what it costs.

Abstract

This paper banks five results from a single investigation and locates, precisely, the point at which the program's derivation chain breaks.

First, a conservation law for interfaces. Defining an interface operationally by two primitives — a partition P of a source and a realization R carrying admitted members across — every property previously proposed as a separate interface role is a derived quantity, and the cost splits into two bills with different algebras. The admission price $A(I) = -\log_2 \mu(\text{Adm})/\mu(S)$ is idempotent under repetition; the crossing price $X(I) = H(X|Y)$ is exactly additive under composition, verified in eight of eight chains to nine decimals. They are not two aspects of one quantity: an interface can separate, constrain, translate and preserve while its crossing price is exactly zero, which is why a type system rejects ill-typed programs and costs nothing at runtime for well-typed ones. Preservation and the crossing price are one measurement read from opposite ends, since $\text{preserve}(I)$ holds if and only if $X(I) = 0$. And the two bills are not independent: together with what reaches the target they conserve the source entropy exactly, $A + X + H(Y) = \log_2 N$, verified as 3.0000 in eighteen configurations with no residual. That converts the question “why this partition” into an optimization with a determinate answer — for a uniform source at fixed capacity, the cost-minimal partition is the balanced one.

Second, the selector is forced rather than imported. The rule “preserve the most future legal paths” converges to the Parry measure as the horizon lengthens (deviation 2.5×10^{-1} to 5.0×10^{-11} at horizons 1 to 64), attaining $\log_2 \lambda$ where four thousand random alternatives never do. The minimal bootable structure requires both a distinction and an alternative: a single self-loop and a two-cycle both have

entropy rate exactly zero, and the first nonzero rate is $\log_2 \phi = 0.694242$. One seam is stated honestly rather than hidden — C1 forbids dead ends, which forces a support constraint, but maximality requires either an information-neutrality principle or an asymptotic reading of C1, and a valid C1 execution at sub-maximal rate exists, so capacity preservation is an independent axiom.

Third, eternal novelty and positive growth rate are independent. The Fibonacci word is deterministic, aperiodic, and mints exactly one new distinguishable state at every step forever — $p(n) = n+1$ — while its asymptotic growth rate is zero. It is a trajectory inside the golden-mean shift, whose capacity is $\log_2 \phi$: the arena has the capacity and the trajectory declines to use it. Separately, the strong no-recurrence condition is shown unsatisfiable on finite deterministic systems — zero of 46,656 maps on one to six states have no periodic point — so it ejects a category rather than filtering within one.

Fourth, a negative result with a machine-precision calibration behind it. The Myrheim–Meyer dimension estimator, calibrated to 0.5% on a two-dimensional sprinkling, is degenerate on the structures the program generates: a three-dimensional grid and a five-dimensional hypercube return 2.2422 and 2.2544, and two genuine two-dimensional grids return 1.8033 and 1.7351 differing only by aspect ratio. The estimator is a monotone reparametrization of comparability density, and comparability density does not determine structure. The values obtained for the program's own posets are therefore uninterpretable — not from finite-size noise, which a matched calibration handles, but from degeneracy.

Fifth, the break in the derivation chain is located and its repair priced. A finite transition system with no terminal states necessarily contains cycles; for a strongly connected system, reachability is not a partial order at all — every pair violates antisymmetry — and path lengths between a state and itself take every value. Condensation by strongly connected components restores acyclicity but, measured on three irreducible systems, collapses the entire graph to a single node, destroying every distinction. The minimal repair that restores a graded order while preserving trajectory distinctions is history unrolling, and it is necessarily infinite. The chain therefore does not derive an interface. It derives why an interface requires a projection from a graded history space onto a compressed state space, and it prices that compression.

Scope of Proof

Domain	Object	Status
Interface algebra	(P,R) generates all roles; A idempotent, X additive	PROVEN
Conservation law	$A + X + H(Y) = \log_2 N$	PROVEN
Optimal partition	Balanced, for a uniform source at fixed capacity	PROVEN
Selector	Parry forced given support + maximality	PROVEN (seam stated)
Capacity as an axiom	Independent of C1; counterexample exhibited	PROVEN

Domain	Object	Status
C3 / C4 independence	Sturmian witness	PROVEN
No-recurrence unsatisfiability	Exhaustive on finite deterministic maps	PROVEN
Myrheim–Meyer degeneracy	Measured against known-dimension controls	PROVEN
Interface break location	Strong connectivity + condensation collapse	PROVEN
Selection rule $\pi: H \rightarrow S$	Not derived	OPEN
Physical readings of any of the above	Structural correspondence only	PULL

1. Method, and Why It Is a Result

The work reported here produced fifteen corrections. Their classification, given in full in Section 8, is the finding with the widest reach, because the dominant class was procedural rather than mathematical. In roughly half the cases a conclusion had been written into the output of a program before the program was run, so that the data arrived pre-interpreted and the contradiction had to be caught by reading one's own text against one's own table. In a further six cases two quantities were compared that were indexed in different spaces — a rendered index against a substrate index, a fibre-size shortcut against a conditional entropy, a strict product order against an inclusion order. In three cases a special case was mistaken for the general law because the test family had been chosen, unwittingly, to make the special case the only one present.

Three procedural rules follow, and they were adopted mid-investigation with measurable effect on the later sections:

Rule 1. Measure before concluding. Print the table; derive afterward. Narration written ahead of execution does not cause errors, but it converts a generation problem into a detection problem, and detection is high but not perfect.

Rule 2. Compare like with like. Before interpreting any numerical comparison, ask whether the two quantities are indexed in the same space. This single question would have prevented the largest identifiable subclass of errors here.

Rule 3. State what the formula computes before stating what it means. A ratio of window counts is a ratio of window counts; whether it deserves to be called efficiency, fidelity, or dimension is a separate claim requiring separate evidence. In this investigation that naming step was wrong at every attempt until the underlying quantity was renamed for what it measured.

2. Result One — The Two Bills and Their Conservation Law

2.1 The operational basis

An interface I over a finite source S and target T is a pair (P, R) : P partitions S with one distinguished cell labelled inadmissible, and R maps the admitted set into T . Write Adm for the admitted set and $R^{-1}(y)$ for the fibre over y . Everything else is derived:

$$\text{constrain}(I) \Leftrightarrow |\text{Adm}| < |S| \quad \text{translate}(I) \Leftrightarrow \exists x \in \text{Adm}: R(x) \neq x$$

$$\text{preserve}(I) \Leftrightarrow |R^{-1}(y)| = 1 \text{ for every } y \text{ in the image}$$

$$A(I) = -\log_2[\mu(\text{Adm})/\mu(S)] \quad X(I) = H(X|Y) = \sum_y p(y) \cdot H(X|Y=y)$$

2.2 The crossing price is not forced

Interface	sep	con	tra	pre	A bits	X bits
Gray recode (bijection)	T	F	T	T	0.0000	0.0000
Quantizer 16 → 4	T	F	T	F	0.0000	2.0000
Validator: even only, injective	T	T	T	T	1.0000	0.0000
Validator + quantizer	T	T	T	F	1.0000	1.0000
Null (identity)	F	F	F	T	0.0000	0.0000

Row three separates, constrains — rejecting half the source — translates, and preserves, with a crossing price of exactly zero. A free crossing exists. Reading the last two columns across every row gives an exact biconditional: $\text{preserve}(I)$ holds if and only if $X(I) = 0$. Preservation and the crossing price are not two properties; they are one measurement from opposite ends.

2.3 Two algebras, identified by their failure modes

$$A(I \circ I) = A(I)$$

idempotent, lattice

$$X(g \circ f) = X(f) + X(g)$$

additive, monoid

Admission idempotence measured at 1.0000/1.0000, 0.0000/0.0000, 2.0000/2.0000. The chain rule holds exactly in eight of eight random chains to nine decimals — for instance $1.272783 + 0.844361 = 2.117144$ against a directly measured 2.117144 — provided $H(Y|Z)$ is computed with Y carrying the pushforward measure. The convenient fibre-size form $\Sigma(|\text{fibre}|/n)\log_2|\text{fibre}|$ is valid only for a uniform source; using it for both terms of the chain produces a false composition law, an error made and corrected during this work.

2.4 The conservation law

$$A(I) + X(I) + H(Y) = \log_2 N$$

Admitted	Cells	Fibre sizes	A	X	H(Y)	Sum
8	2	(4,4)	0.0000	2.0000	1.0000	3.0000
8	2	(1,7)	0.0000	2.4564	0.5436	3.0000
8	3	(3,3,2)	0.0000	1.4387	1.5613	3.0000
6	3	(2,2,2)	0.4150	1.0000	1.5850	3.0000
6	2	(1,5)	0.4150	1.9349	0.6500	3.0000
4	3	(1,1,2)	1.0000	0.5000	1.5000	3.0000

Eighteen configurations, sum 3.0000 in every one, no residual. The source carries $\log_2 N$ bits; admission destroys A by refusing entry; realization destroys X by merging what it admits; H(Y) arrives. Minimizing total destruction is therefore identical to maximizing delivered information, and at fixed capacity the maximizer is the balanced partition: (4,4) at capacity two, (3,3,2) at three, (2,2,2,2) at four. The partition is not chosen; given a capacity it is forced. This is a uniform-source result — for a non-uniform source the criterion becomes equal probability mass rather than equal count — but the conservation law itself follows from the entropy chain rule and holds for any source measure.

3. Result Two — The Selector Is Forced, With One Seam

If the compiler is the closure of the transition system rather than an object above it, a selector cannot be imported without regress. The candidate rule is that a transition should preserve the most future legal paths, since a transition into a dead end violates C1. Measured, the rule converges to the Parry measure:

Horizon	$\max P_{\text{greedy}} - P_{\text{Parry}} $	Entropy rate
1	2.500×10^{-1}	0.44284169
4	1.093×10^{-3}	0.55137923
16	3.331×10^{-7}	0.55146309
64	4.996×10^{-11}	0.55146309

Parry attains $\log_2 \lambda = 0.55146309$ and four thousand random alternative measures never exceed it. (Classical: Parry 1964, measure of maximal entropy. Instrument, marked.)

3.1 The boot condition

Structure	Entropy rate	Executable?
One state, self-loop	0.000000	No — change without distinction
Two-cycle	0.000000	No — distinction without alternative

Structure	Entropy rate	Executable?
Two states, one branch	0.694242	Yes

Neither ingredient suffices alone. The minimum requires a distinction and an alternative, and the minimal structure is the golden-mean shift with rate $\log_2 \phi = 0.694242$ — ϕ being the smallest Perron root above one for a binary adjacency matrix, so this follows from minimality of the constraint rather than from anything numerological.

3.2 The seam, stated

C1 forbids dead ends, which forces a support constraint: never take a transition into a state with no future. Maximizing entropy is strictly stronger than avoiding zero. The gap can be closed two ways, and both are additional to C1 as stated at the transition level. Either one adds an information-neutrality principle — a selector that adds information smuggles in a hidden primitive — or one reads C1 asymptotically, on the path measure rather than transition by transition, in which case a sub-maximal measure lives on an exponentially vanishing fraction of the legal paths and has, in the limit, abandoned almost every future the constraint left open. Measured: at depth 100 the uniform-on-legal selector retains 3.09×10^{-3} of the legal path set, an arbitrary skewed selector 3.91×10^{-34} , and the deterministic greedy-argmax selector 7.28×10^{-52} — the rule that sounds like it preserves possibility destroys it fastest, because it is deterministic and walks one path out of λ^n .

The bootloader test settles the status of capacity as an axiom. A deterministic execution fails C1 on a finite arena, since pigeonhole forces it into a cycle and a cycle is a disguised fixed point. But uniform-on-legal is a perfectly valid C1 execution — every state has a successor, the process is aperiodic, entropy rate $1.615370 > 0$ — while sitting 0.083396 bits per step below maximum. So C1 forbids capacity zero and does not force capacity maximum. Capacity preservation is an independent axiom, not a derived consequence.

4. Result Three — Novelty and Capacity Are Independent

The Fibonacci word, generated by the substitution $a \rightarrow ab$, $b \rightarrow a$, is deterministic and aperiodic — no period below 3000 across 20,000 symbols — with factor complexity $p(n) = n+1$ exactly: 2, 3, 4, 6, 9, 13, 17, 21 at $n = 1, 2, 3, 5, 8, 12, 16, 20$, and a difference sequence that is a flat run of ones. It mints exactly one new distinguishable state at every step, forever. Its growth rate is zero: $(1/n)\log_2 p(n)$ decays 0.517, 0.346, 0.220, 0.134, 0.079, 0.046.

So a system can generate genuine novelty at every step for eternity and have no asymptotic capacity. Novelty never stalls; it never accelerates. The two conditions are independent, and the witness is explicit.

The witness and its arena are the same ϕ wearing two hats. The Fibonacci word lives inside the golden-mean shift — b is always followed by a , so no two consecutive b 's — and that arena has capacity $\log_2 \phi$.

The arena has the capacity; the trajectory declines to use it. It is a maximally non-Parry trajectory: deterministic, walking one path out of ϕ^n . This is the same distinction as Section 3.2 at its extreme.

4.1 The no-recurrence condition is unsatisfiable, not demanding

Requiring $\text{Fix}(f^n) = \emptyset$ for every n is not a strong constraint on finite deterministic systems; it is an impossible one. Exhaustively, across every map on one to six states — 46,656 maps at $n = 6$ — exactly zero have no periodic point. Pigeonhole guarantees it. The condition therefore ejects a category rather than filtering within one, and the three available escapes are an infinite arena, non-determinism, or a generated arena. The first and third are the same escape in two languages: a Sturmian sequence is the symbolic coding of an irrational rotation, so “infinite state space” and “growing library of distinguishable observations” describe one object.

An earlier claim in this program that C1 forces non-determinism is retracted here. It forces the failure of at least one of four assumptions — finiteness, determinism, a time-independent map, or a pre-existing state space — and non-determinism is only one of them.

5. Result Four — Order Dimension Does Not Read Off the Structures

The Myrheim–Meyer estimator inverts the ratio of comparable pairs to squared cardinality against a known function of dimension. Calibrated on a two-dimensional sprinkling, where the light-cone coordinates make a causal diamond into the unit square with componentwise order, it recovers $d = 2.0103$ against a predicted ordinate of 0.2500 — 0.5% error, the exact case.

Applied to structures with known order dimension, under the relation those structures actually carry, it is degenerate:

Structure	N	$L = C_2/N^2$	MM d	True order dimension
(5,5,5) grid	125	0.208000	2.2422	3
(2,2,2,2,2) hypercube	32	0.206055	2.2544	5
(11,11) grid	121	0.289256	1.8033	2
(25,5) grid	125	0.304000	1.7351	2
Boolean lattice 2^7	128	0.125671	2.8822	7

A three-dimensional grid and a five-dimensional hypercube agree to two decimals. Two genuine two-dimensional grids disagree by 0.07, differing only in aspect ratio. The estimator is a monotone reparametrization of comparability density, and many shapes share a density. The closed form for products of chains reproduces both controls to machine precision (0.208000 and 0.125671), so this is not a numerical artifact.

Consequently the values obtained for the program's own structures — Fibonacci factor poset 1.8778, prefix-closed language lattice 1.8789, subspace lattice of $\text{GF}(2)^4$ 3.1723 — are uninterpretable as

dimensions. They are comparability densities of 0.2738, 0.2736 and 0.0994. The first two agreeing to three decimals despite different objects at different sizes is a coincidence of density, not structural agreement.

The negative result is the useful part, and it generalizes. A single scalar summarizing an order cannot discriminate structures that differ in shape at equal density. Any programme intending to read geometry off a bare relation needs a family of invariants — interval structure, chain and antichain profiles, spectral data — or a family-specific calibration. This mirrors the field's own experience: a single dimension estimator proved insufficient and the response was more chain-abundance statistics, not a reinterpretation of the original scalar.

6. Result Five — The Break, Located and Priced

The program's derivation chain runs: C1 forces transitions; transitions force adjacency; adjacency forces a topology; topology forces distinction. It then proposes an interface. That step does not follow, and the reason is now specific.

6.1 The transition structure carries no depth

C1 on a finite arena forbids terminal states, so by pigeonhole every trajectory enters a cycle. For a strongly connected system the consequence is severe: reachability is not a partial order at all. On the golden-mean transition graph, all four pairs — (a,a), (a,b), (b,a), (b,b) — violate antisymmetry, and path lengths from a back to a take every value in {1,2,3,4,5,6,7,8}. Depth is not ambiguous; it is undefined.

6.2 The cheap repair destroys everything it orders

Condensation by strongly connected components restores acyclicity and therefore a partial order, and requires no infinity. Measured on three irreducible systems, it also destroys every distinction:

Transition system	States	SCCs	Condensation nodes	Distinctions surviving
Golden-mean shift	2	1	1	1
Full shift on {a,b}	2	1	1	1
Fibonacci 2-block graph	3	1	1	1
Control: chain with sink loop (reducible)	3	3	3	3

For an irreducible system the condensation is a single point. It solves antisymmetry by deleting the system. The reducible control retains all three nodes, confirming the collapse is a property of strong connectivity rather than of the method.

6.3 The repair that works, and its price

History unrolling — states are admissible words, order is prefix — restores both a genuine order and a rank. Measured on the Fibonacci language: zero cover relations violating rank + 1 across 44 admissible

words, and every word carries a unique depth. The rank function $r(s_0, \dots, s_n) = n$ exists because every cover adds exactly one transition.

It should be recorded that the two posets on which the estimator of Section 5 was run — the factor poset and the prefix-closed language lattice — are themselves graded, with zero cover violations, by length and by cardinality respectively. Their lattice-like behaviour under that estimator therefore needs no further explanation, and testing their gradedness is not a test. The non-trivial question was always whether the transition structure is graded, and it is not, because it is not an order.

The precise statement, with the cheaper alternative ruled out rather than left implicit: for a strongly connected finite transition system with no terminal states, recurrence destroys any intrinsic state-space ordering. Condensation restores a finite partial order but collapses the irreducible component to a single class, removing the distinctions depth would measure. The minimal structure that preserves trajectory distinctions while restoring a graded order is history unrolling, and it is necessarily infinite.

A finite state description cannot carry temporal depth. Temporal depth exists only in trajectory space.

This does not derive an interface. It derives that an interface requires a projection π from a graded history space H onto a compressed state space S , and prices the compression: finite state description is bought with infinite temporal resolution, or temporal resolution is bought with an infinite arena. That trade is the conservation law of Section 2 appearing at a different seam — many histories collapse onto one observed state, and what the projection destroys is not bits but branching, the paths not taken.

The remaining open question is correspondingly narrow and, for the first time in this chain, isolated rather than hidden inside a derivation step: what selects π ?

7. The Chain, As It Now Stands

Step	Status
$C1 \Rightarrow$ transitions exist	Forced — one change is minimally an ordered pair (before, after)
Transitions $\Rightarrow$ adjacency	Forced — adjacency is which transitions exist
Adjacency $\Rightarrow$ topology	Forced — reachability, components, cycles
Topology $\Rightarrow$ distinction	Forced — reachable versus not partitions the graph
Distinction $\Rightarrow$ interface	NOT FORCED — requires a depth coordinate the transition structure lacks
... via history unrolling	Available, graded, and necessarily infinite
... via condensation	Available, finite, and destroys all distinctions in the irreducible case
Interface $\Rightarrow$ measurement, laws, ...	Not attempted; the chain is broken above it

8. Corrections, Classified

Fifteen corrections were made. The classification matters more than the list, because it identifies which are mathematical and which are procedural.

Class	Count	Examples
Narration written before execution	6	“the two costs move independently” with three of four quadrants produced; “LATTICE IS FORCED” printed above a test returning False; “alpha verified” above a failed chain-rule test
Frame mismatch (indices in different spaces)	4	$p_R(n)$ against $p_S(n)$ rather than $p_S(W(n))$, twice; $H(X Z)$ computed twice under two names; a strict product order compared against an inclusion order
Special case mistaken for general law	3	Bhattacharyya for Chernoff; a binary partition of operations where four classes exist; a tautology reported as an empirical coincidence
Scope overreach	2	“C1 forces non-determinism”; “no additive valuation unless graded”

Two corrections deserve individual note because they changed findings rather than wording. Keying a trade-off table on an image-size proxy rather than the true conditional entropy conflated two fibre partitions and produced four buckets where five exist; rekeying strengthened the result. And reporting that transition-level admissibility separates all dynamics was definitional, not empirical: the admissible transition set of a deterministic system is the graph of its map.

Status Ledger

Claim	Status	Basis
$\text{preserve}(I) \Leftrightarrow X(I) = 0$	PROVEN	Exact biconditional, five test interfaces
A idempotent; X additive under composition	PROVEN	1.0000/1.0000 etc.; 8 of 8 chains to 9 decimals
$A + X + H(Y) = \log_2 N$	PROVEN	18 configurations, 3.0000 exactly, no residual
Balanced partition is cost-minimal (uniform source)	PROVEN	(4,4), (3,3,2), (2,2,2,2) optimal at capacities 2,3,4
Free crossings exist	PROVEN	Validator: $A = 1.0000$, $X = 0.0000$, all four roles hold
Greedy reachability $\rightarrow$ Parry	PROVEN	$2.5e-1 \rightarrow 5.0e-11$ at horizons $1 \rightarrow 64$
Capacity preservation independent of C1	PROVEN	Uniform-on-legal: valid C1 execution, 0.0834 bits below max
Boot minimum = distinction + alternative	PROVEN	0, 0, then $\log_2 \phi = 0.694242$
C3 and C4 independent	PROVEN	Fibonacci word: $p(n) = n+1$, rate $\rightarrow 0$
$\text{Fix}(f^n) = \emptyset$ unsatisfiable on finite deterministic	PROVEN	0 of 46,656 maps at $n = 6$

Claim	Status	Basis
MM degenerate on these structures	PROVEN	3-D grid 2.2422 vs 5-D cube 2.2544; controls to machine precision
Reachability not a partial order (irreducible case)	PROVEN	4 of 4 antisymmetry violations; path lengths {1..8}
Condensation collapses irreducible systems to a point	PROVEN	3 systems, 1 node each; reducible control retains 3
History unrolling graded	PROVEN	0 cover violations, unique depth, 44 words
Selection rule $\pi: H \rightarrow S$	OPEN	Isolated, not hidden
Non-uniform-source optimum	OPEN	Criterion becomes equal mass; unmeasured
Interface derived	NOT CLAIMED	Only its requirement and its price
Physical instantiation of any result here	PULL	Structural correspondence only

Appendix: Live Output, Quoted Verbatim

```

CONSERVATION  A + X + H(Y) = log2(N) = 3.0000
  admitted cells      fibre      A      X      H(Y)      sum
      8      2      (4, 4)  -0.0000  2.0000  1.0000  3.0000
      8      3      (3, 3, 2) -0.0000  1.4387  1.5613  3.0000
      6      3      (2, 2, 2)  0.4150  1.0000  1.5850  3.0000
      4      3      (1, 1, 2)  1.0000  0.5000  1.5000  3.0000

CHAIN RULE      X(f)      X(g|pushforward)      sum      H(X|Z)      match
      1.272783      0.844361  2.117144  2.117144  True
      1.647783      1.154025  2.801808  2.801808  True

GREEDY -> PARRY      horizon      max|P_greedy - P_Parry|      entropy rate
                        1                        2.500e-01      0.44284169
                        16                       3.331e-07      0.55146309
                        64                       4.996e-11      0.55146309

BOOT      one state self-loop 0.000000 | two-cycle 0.000000
      two states + branch 0.694242 = log2(phi)

ABANDONMENT at depth 100: Parry 1.000 | uniform 3.09e-03
      skewed 3.91e-34 | greedy-argmax (deterministic) 7.28e-52

STURMIAN  p(n) = 2,3,4,6,9,13,17,21 at n=1,2,3,5,8,12,16,20 (= n+1)
      new states per step: [1,1,1,1,1,1,1,1,1,1]
      (1/n)log2 p(n): 0.517 0.346 0.220 0.134 0.079 0.046 -> 0
      smallest period up to 3000 over 20000 symbols: None

NO-RECURRENCE  n      maps n^n      maps with NO periodic point
      4      256      0
      5      3125      0
      6      46656      0

```

MYRHEIM-MEYER (relation matched: $\leq$ in all coords, at least one strict)

grid	N	L	MM d	true dim
(125,)	125	0.496000	1.0131	1
(11, 11)	121	0.289256	1.8033	2
(25, 5)	125	0.304000	1.7351	2
(5, 5, 5)	125	0.208000	2.2422	3
(2,2,2,2,2)	32	0.206055	2.2544	5
(2,)*7	128	0.125671	2.8822	7

closed form vs measured divisor poset : match=True
closed form vs measured Boolean 2^7 : match=True

GRADEDNESS factor poset ($|F|=35$): cover violations = 0
prefix-closed lattice ($|L|=25$): cover violations = 0
admissible-word poset ($|W|=44$): cover violations = 0
every word has a unique depth: True

TRANSITION STRUCTURE {'a': ['a','b'], 'b': ['a']}
reachability: a->['a','b'], b->['a','b']
antisymmetry violations: [('a','a'),('a','b'),('b','a'),('b','b')]
is reachability a partial order? False
path lengths from 'a' to 'a': [1, 2, 3, 4, 5, 6, 7, 8]

CONDENSATION golden-mean SCCs=1 condensation nodes=1
full shift SCCs=1 condensation nodes=1
Fibonacci 2-block SCCs=1 condensation nodes=1
reducible control SCCs=3 condensation nodes=3

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